

Total No. of Questions : 8]

SEAT No. :

P9294

[Total No. of Pages : 2

[6181]520

**B.E. (Computer Engineering)
INFORMATION AND CYBER SECURITY
(2015 Pattern) (Semester - II) (410251)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data, if necessary.

- Q1)** a) Define basic terminology in network Security? Differentiate between confidentiality and authentication. **[8]**
- b) Demonstrate Diffie-Hellman Key exchange algorithm? **[6]**
- c) Identify use of X. 509 Certificate with diagram. **[6]**

OR

- Q2)** a) Illustrate the terms **[8]**
- i) Public Key/ Private Key
 - ii) Privacy Vs Security
 - iii) Plaintext
 - iv) Ciphertext
- b) Demonstrate Data Encryption Standard Algorithm? **[6]**
- c) Define RSA, Illustrate your answer with Example. **[6]**

- Q3)** a) Define modes of operation of IPSEC with diagram. **[8]**
- b) Demonstrate SSL protocol architecture in detail. **[8]**

OR

- Q4)** a) Illustrate following Terms with suitable diagram. **[8]**
- i) Change cipher spec protocol
 - ii) Handshake Protocol
 - iii) Alert Protocol
- b) Illustrate IP sec Transport and Tunnel Mode? Explain VPN. **[8]**

P.T.O.

Q5) a) What is Intrusion detection System? Explain types based on method of detection. [9]

b) Define and Explain Firewall, its architecture, limitation and applications. [8]

OR

Q6) a) Explain types of IDS, its limitation and challenges. [9]

b) Illustrate password management in detail. [8]

Q7) a) Define following terms in detail. [9]

i) Nmap

ii) Snork

iii) Wireshark

b) What is SQL injection attack? How to protect data from SQL Injection? [8]

OR

Q8) a) Write short note on Indian IT Act 2000 and amendments in 2008? [9]

b) Outline the Phishing concepts? What are the techniques used by phishers to launch it? [8]